

A Technical Introduction to Explainable artificial intelligence

Section 1

Technical complexity A fundamental barrier to making AI systems explainable is the technical complexity of such systems. End users often lack the coding knowledge required to understand software of any kind. Current methods used to explain AI are mainly technical ones, geared toward machine learning engineers for debugging purposes, rather than toward the end users who are ultimately affected by the system, causing "a gap between explainability in practice and the goal of transparency". Proposed solutions to address the issue of technical complexity include either promoting the coding education of the general public so technical explanations would be more accessible to end users, or providing explanations in layperson terms. The solution must avoid oversimplification. It is important to strike a balance between accuracy – how faithfully the explanation reflects the process of the AI system – and explainability – how well end users understand the process. This is a difficult balance to strike, since the complexity of machine learning makes it difficult for even ML engineers to fully understand, let alone non-experts.

Section 2

Some explainability techniques don't involve understanding how the model works, and may work across various AI systems. Treating the model as a black box and analyzing how marginal changes to the inputs affect the result sometimes provides a sufficient explanation.

Section 3

Within artificial intelligence (AI), explainable AI (XAI), generally overlapping with interpretable AI or explainable machine learning (XML), is a field of research that explores methods that provide humans with the ability of intellectual oversight over AI algorithms. The main focus is on the reasoning behind the decisions or predictions made by the AI algorithms, to make them more understandable and transparent. This addresses users' requirement to assess safety and scrutinize the automated decision making in applications. XAI counters the "black box" tendency of machine learning, where even the AI's designers cannot explain why it arrived at a specific decision. XAI hopes to help users of AI-powered systems perform more effectively by improving their understanding of how those systems reason. XAI may be an implementation of the social right to explanation. Even if there is no such legal right or regulatory requirement, XAI can improve the user experience of a product or service by helping end users trust that the AI is making good decisions. XAI aims to explain what has been done, what is being done, and what

will be done next, and to unveil which information these actions are based on. This makes it possible to confirm existing knowledge, challenge existing knowledge, and generate new assumptions.

Section 4

Partial dependency plots show the marginal effect of an input feature on the predicted outcome. SHAP (SHapley Additive exPlanations) enables visualization of the contribution of each input feature to the output. It works by calculating Shapley values, which measure the average marginal contribution of a feature across all possible combinations of features. Feature importance estimates how important a feature is for the model. It is usually done using permutation importance, which measures the performance decrease when the feature value is randomly shuffled across all samples. LIME (Local Interpretable Model-Agnostic Explanations method) approximates locally a model's outputs with a simpler, interpretable model. Multitask learning provides a large number of outputs in addition to the target classification. These other outputs can help developers deduce what the network has learned. For images, saliency maps highlight the parts of an image that most influenced the result. Systems that are expert or knowledge based are software systems that are made by experts. This system consists of a knowledge based encoding for the domain knowledge. This system is usually modeled as production rules, and someone uses this knowledge base which the user can question the system for knowledge. In expert systems, the language and explanations are understood with an explanation for the reasoning or a problem solving activity. However, these techniques are not very suitable for language models like generative pretrained transformers. Since these models generate language, they can provide an explanation, but which may not be reliable. Other techniques include attention analysis (examining how the model focuses on different parts of the input), probing methods (testing what information is captured in the model's representations), causal tracing (tracing the flow of information through the model) and circuit discovery (identifying specific subnetworks responsible for certain behaviors). Explainability research in this area overlaps significantly with interpretability and alignment research.

Section 5

Payoff allocation Nizri, Azaria and Hazon present an algorithm for computing explanations for the Shapley value. Given a coalitional game, their algorithm decomposes it to sub-games, for which it is easy to generate verbal explanations based on the axioms characterizing the Shapley value. The payoff allocation for each sub-game is perceived as fair, so the Shapley-based payoff allocation for the given game should seem fair as well. An experiment with 210 human subjects shows that, with their automatically generated explanations, subjects perceive

Shapley-based payoff allocation as significantly fairer than with a general standard explanation.

Section 6

History and methods During the 1970s to 1990s, symbolic reasoning systems, such as MYCIN, GUIDON, SOPHIE, and PROTOS could represent, reason about, and explain their reasoning for diagnostic, instructional, or machine-learning (explanation-based learning) purposes. MYCIN, developed in the early 1970s as a research prototype for diagnosing bacteremia infections of the bloodstream, could explain which of its hand-coded rules contributed to a diagnosis in a specific case. Research in intelligent tutoring systems resulted in developing systems such as SOPHIE that could act as an "articulate expert", explaining problem-solving strategy at a level the student could understand, so they would know what action to take next. For instance, SOPHIE could explain the qualitative reasoning behind its electronics troubleshooting, even though it ultimately relied on the SPICE circuit simulator. Similarly, GUIDON added tutorial rules to supplement MYCIN's domain-level rules so it could explain the strategy for medical diagnosis. Symbolic approaches to machine learning relying on explanation-based learning, such as PROTOS, made use of explicit representations of explanations expressed in a dedicated explanation language, both to explain their actions and to acquire new knowledge. In the 1980s through the early 1990s, truth maintenance systems (TMS) extended the capabilities of causal-reasoning, rule-based, and logic-based inference systems. A TMS explicitly tracks alternate lines of reasoning, justifications for conclusions, and lines of reasoning that lead to contradictions, allowing future reasoning to avoid these dead ends. To provide an explanation, they trace reasoning from conclusions to assumptions through rule operations or logical inferences, allowing explanations to be generated from the reasoning traces. As an example, consider a rule-based problem solver with just a few rules about Socrates that concludes he has died from poison: